

RAS AL KHAIMAH INTL, United Arab Emirates

WMO: 411840

Lat: 25.613N

Lon: 55.939E

Elev: 31

StdP: 100.95

Time Zone: 4.00 (E04)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	10.3	11.8	1.9	4.4	26.4	4.0	5.1	25.4	8.5	23.0	7.5	23.2	1.1	210	0.400

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.1	44.7	24.4	43.5	24.9	42.6	25.2	30.7	37.9	30.0	37.2	29.3	36.8	4.8	330

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
29.1	25.9	34.7	28.2	24.5	34.5	27.7	23.7	34.4	104.2	38.2	100.4	37.3	97.3	36.7	36.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)
8.2	7.1	6.2	DB	7.6	47.0	1.4	0.9	6.6	47.6	5.8	48.1	5.0	48.6	4.0	49.2	4.0
			WB	5.5	32.4	1.7	1.7	4.3	33.6	3.2	34.6	2.2	35.6	1.0	36.8	1.0

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	28.4	18.9	20.4	23.5	28.0	32.5	34.8	36.6	36.2	33.6	29.7	24.8	20.6
	DBStd	6.55	2.00	2.59	2.68	2.68	2.05	1.55	1.37	1.32	1.51	1.76	2.18	2.13
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	28	17	7	0	0	0	0	0	0	0	0	0	4
	CDD10.0	6697	276	292	419	539	697	744	824	812	708	612	446	329
	CDD18.3	3684	34	66	161	289	438	494	566	554	458	353	196	76
	CDH23.3	54613	268	534	1558	3628	6716	8204	9724	9446	7184	4716	2011	623
Wind	WSAvg	2.4	2.0	2.4	2.5	2.5	2.7	2.8	3.0	2.9	2.4	2.1	2.0	1.9
	PrecAvg	115	23	25	31	4	1	0	2	0	0	3	5	19
	PrecMax	303	111	146	160	27	2	2	28	4	5	41	61	152
	PrecMin	28	0	0	0	0	0	0	0	0	0	0	0	0
	PrecStd	80	28	30	38	7	0	0	5	1	1	8	12	35
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	29.0	32.8	36.8	41.0	44.1	45.9	46.0	45.3	43.3	40.0	35.2	31.0
		MCWB	18.5	19.0	19.9	21.6	23.3	23.6	23.9	24.4	24.0	24.0	21.3	20.1
	2%	DB	27.3	29.9	34.2	39.0	42.7	44.1	44.4	44.1	42.0	38.7	33.9	29.2
		MCWB	18.3	19.0	19.4	21.1	22.9	24.1	24.9	25.2	24.4	23.7	21.5	19.7
	5%	DB	26.1	28.1	32.1	37.5	41.2	42.9	43.2	43.0	40.8	37.2	32.8	28.0
		MCWB	18.2	18.7	19.4	20.8	22.9	24.8	25.7	25.7	24.8	23.7	22.0	19.6
	10%	DB	24.9	26.6	30.2	35.8	40.0	41.8	42.1	41.9	39.8	36.1	31.2	26.9
		MCWB	17.8	18.4	19.3	20.8	23.2	25.1	26.2	26.2	25.4	23.8	21.8	19.2
	0.4%	WB	21.0	22.3	23.4	24.9	28.7	30.6	32.2	31.5	30.7	29.1	25.4	22.7
		MCDB	25.7	28.1	30.4	33.4	37.7	38.8	39.3	38.4	36.6	34.5	31.0	27.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	20.1	21.1	22.3	23.8	27.1	29.6	30.9	30.4	29.7	28.0	24.4	21.7
		MCDB	24.9	26.2	28.8	32.7	36.4	37.4	38.4	37.4	35.9	33.3	30.2	26.7
	5%	WB	19.4	20.2	21.5	23.0	26.1	28.8	30.1	29.7	29.0	27.1	23.7	20.9
		MCDB	24.1	25.5	27.5	32.2	35.7	36.9	37.7	37.0	35.2	32.7	29.6	26.1
	10%	WB	18.5	19.5	20.7	22.3	25.1	28.0	29.4	29.0	28.3	26.3	22.8	20.1
		MCDB	23.3	24.7	26.6	31.2	35.2	36.2	37.2	37.0	34.6	32.2	29.0	25.4
Mean Daily Temperature Range	5% DB	MDBR	11.9	12.1	12.7	14.6	15.5	14.5	12.1	12.1	13.4	14.3	13.1	12.3
		MCDBR	13.6	14.0	16.0	17.3	17.1	16.4	13.9	14.1	15.0	15.1	14.3	13.9
		MCWBR	6.8	6.5	6.5	6.3	6.7	6.8	6.1	6.2	6.3	6.6	6.7	7.1
	5% WB	MCDBR	11.7	11.6	12.3	14.4	15.3	13.7	11.2	11.4	12.8	13.7	13.3	12.3
		MCWBR	6.4	6.1	5.7	6.3	7.1	6.7	5.7	5.8	5.8	6.2	6.7	6.7
Clear-Sky Solar Irradiance	taub		0.449	0.514	0.575	0.609	0.607	0.664	0.750	0.708	0.619	0.533	0.467	0.448
	taud		2.091	1.912	1.770	1.720	1.725	1.653	1.549	1.620	1.748	1.904	2.070	2.094
	Ebn at Noon		786	761	739	726	726	681	624	648	694	733	762	767
	Edh at Noon		143	183	222	239	237	253	281	261	224	182	145	137
All-Sky Solar Radiation	RadAvg		3.96	4.74	5.66	6.56	7.36	7.48	6.97	6.59	6.11	5.34	4.30	3.75
	RadStd		0.28	0.38	0.40	0.40	0.17	0.16	0.18	0.18	0.09	0.09	0.13	0.25

Historical Trends

		Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)
Station Only	N/A	N/A	-1.96	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (15 neighbors)	+0.41	+0.60	N/A	N/A	N/A	N/A	N/A	N/A	+130	+125

Nomenclature: See separate page